AAROHA TECHNOLOGIES | ACADEMY LMS

STATUS: PRODUCTION LIVE

Full Day Engineering & Feature Development Summary Sheet

Date: November 12, 2026
Live Target: lms.aaroha-inc.com
Release Branch: main (eb7c51e)

Total Commits	CI/CD Deploys	Vite Build Time	Core Views	Production Uptime
19	100%	513 ms	5 Distinct	HTTP 200
Production Pushes	Automated Success	Optimized Bundle	Catalog, Brochure, Player, Dashboard, Admin	Hostinger LAMP Live

1. Executive Summary & Architecture Modernization

On November 12, 2026, the Aaroha Learning Management System (LMS) underwent comprehensive full-stack modernizations, transforming from an initial prototype into a high-performance, Coursera-calibrated executive education platform. Key advancements spanned full-stack architecture stabilization, automated CI/CD deployment pipeline engineering with SHA-256 delta caching, brand identity contrast upgrades, continuous media embed engine normalizations, and an enterprise access-control gate separating public brochures from immersive video classrooms.

2. Detailed Engineering Milestones & Key Developments

Milestone 1: Platform Foundation & Full-Stack Architecture

Commit e1270f0 · Architecture baseline

- Constructed modern React 19 + Vite 6 frontend styled with clean Tailwind CSS utility architecture.
- Engineered production PHP 8.x REST adapter service with modular controller endpoints and JWT authentication.
- Designed relational MySQL 8 schema (database/schema.sql) supporting users, roles (student/admin), courses, modules, lessons, task defense submissions, and completion tracking.
- Replaced confusing theoretical 'Curricula' terminology with executive 'Courses & Specializations'.

Milestone 2: High-Performance CI/CD Automated Deployment Engine

Commits a1ee2d6 -> 46e0c4d · Hostinger LAMP Pipeline

- Replaced error-prone third-party FTP GitHub Actions with a resilient native Python FTP synchronization engine (.github/scripts/deploy.py).
- Automated path sanitization and normalization for Hostinger root: domains/aaroha-inc.com/public_html/lms/.
- Resolved FTPS passive socket reset issues with streamlined FTP protocol configurations.
- Implemented SHA-256 hash-based delta caching via .deploy_manifest.json, reducing deployment sync duration from 26 seconds to 16 seconds.

Milestone 3: Dynamic Faculty Console & Interactive Controls

Commit 5f106db · Faculty Director Portal

- Fixed category selection locking; implemented interactive two-way pill toggle ('Existing List' vs '+ Add New Domain').
- Integrated inline restore button (RotateCcw) allowing instant return to existing catalog domains.
- Added multi-module course creator with real-time syllabus definition, executive summaries, target seniority levels, and durations.

Milestone 4: State Hydration, Auth Persistence & Zero-Logout Reloads

Commit 5368e0f · Client Storage Architecture

- Eliminated session dropouts on browser page refresh by hydrating user authentication state, JWT tokens, active view, and selected course ID from localStorage on root mount.
- Established strict role-based routing between Faculty Director consoles and Learner workspaces.

Milestone 5: Video Normalization Engine & Media Security

Commit 296f1f4 · videoUtils.js Engine

- Engineered video embed normalizer parsing YouTube standard watch URLs, shorts, and youtu.be links into privacy-compliant youtube-nocookie.com/embed endpoints.
- Overcame X-Frame-Options: SAMEORIGIN 'refused to connect' iframe blocking.
- Built automated 16:9 YouTube thumbnail fetcher (img.youtube.com/vi/{id}/mqdefault.jpg) with high-resolution fallback.

Milestone 6: Brand Identity, Contrast Calibration & Favicons

Commits 219ce82, c39abbc, d317075 · Visual Brand Standards

- Recalibrated official Aaroha logo (aaroha_logo.webp) by transforming light tagline text to dark slate #0f172a, ensuring high contrast against pure white backgrounds.
- Wrapped logo containers in clean white badge pills (bg-white border-slate-200 shadow-xs) everywhere across navigation, hero cards, course thumbnails, modals, and footers.
- Replaced Vite template icons with official Aaroha electric blue peak emblem on white rounded square containers (both vector favicon.svg and multi-size binary favicon.ico).

Milestone 7: Dark Cinema Classroom / Course Interface

Commit 389ef27 · CourseClassroomView.jsx

- Constructed full-bleed dark navy (#0d1b2a) cinema-mode classroom matching LearnTube / Coursera reference designs.
- Integrated dedicated header with [Back To Home] button and centered white Aaroha Academy logo badge.
- Built interactive quick-actions bar: dynamic thumbs up/like tracking, view counter, report action, and duration pill.
- Designed continuous vertical timeline playlist with checkmark nodes, active playback rings, and lesson thumbnail previews.
- Built multi-tab content space for Lecture Notes, Hands-On Evaluated Assignments, and Faculty Accreditation.

Milestone 8: Decoupled 'My Learning' Enrolled Programs Dashboard

Commit eb3a58c · StudentPortalView.jsx

- Decoupled My Learning navigation: clicking My Learning now displays enrolled courses first instead of jumping directly into video playback.
- Created KPI metric tiles: Enrolled Programs, In Progress, Completed Lectures, Accreditations Issued.
- Built card grid featuring real-time green progress bars (X% Completed), Next Up lecture prompts, and [Resume Course] action buttons.

Milestone 9: Strict Role-Based Access Gating & Course Brochure

Commit eb7c51e · CourseBrochureView.jsx & Access Security

- Implemented strict access control: un-enrolled learners are restricted exclusively to the Course Brochure (CourseBrochureView).
- Course Brochure features program highlights, syllabus breakdown with lock badges, faculty director profile, and enrollment CTA.
- Only enrolled programs unlock the Course Interface (CourseClassroomView). Enrolling instantly unlocks the live video player and learning workspace.

3. Production File Registry & Component Taxonomy

The following table documents the active source files, their operational roles, and current implementation states across the Aaroha LMS repository.

Component / File Path	Operational Role	Target Audience / Gating	Status
frontend/src/App.jsx	Root router, auth session hydration, view transitions, and global state coordinator.	Global / Root container	**Active · Verified**
frontend/src/components/UserCatalogView.jsx	Public catalog grid, topic filters, accreditation banners, and course discovery cards.	All Visitors & Learners	**Active · Verified**
frontend/src/components/CourseBrochureView.jsx	Public course landing page, syllabus overview with locked lectures, and enrollment CTA.	Un-enrolled Learners	**Active · Verified**
frontend/src/components/CourseClassroomView.jsx	Full-bleed dark cinema classroom, video player, timeline playlist, and task defense.	Enrolled Candidates Only	**Active · Gated**
frontend/src/components/StudentPortalView.jsx	My Learning dashboard, enrolled courses grid, progress bars, and KPI metric tiles.	Enrolled Students	**Active · Verified**
frontend/src/components/AdminPortalView.jsx	Faculty director console, course creation with domain pills, and task defense grading.	Faculty Admin Only	**Active · Role-Gated**
frontend/src/videoUtils.js	Normalizes YouTube/Vimeo URLs to embed format and extracts 16:9 thumbnail previews.	Internal Media Engine	**Active · Verified**
frontend/public/favicon.svg & .ico	Official Aaroha electric blue peak emblem on clean white rounded background.	Browser Tab & OS Bookmarks	**Active · 200 OK**
.github/scripts/deploy.py	Native Python FTP deployment engine with SHA-256 delta caching for Hostinger.	CI/CD Automation	**Active · 16s Sync**

4. Complete Git Commit Audit Trail (November 12, 2026)

Commit Hash	Scope & Commit Description	Key Impact & Verification
eb7c51e	feat(access): restrict course interface to enrolled courses	Un-enrolled users restricted to brochure; classroom gated.
eb3a58c	feat(student): show enrolled courses dashboard on my learning	My Learning shows progress cards first without jumping to video.
389ef27	feat(classroom): redesign course detail into full-bleed cinema classroom	Dark navy (#0d1b2a), vertical timeline playlist, quick actions.
d317075	feat(favicon): replace vite icon with official aaroha emblem on white bg	Generated vector SVG and multi-resolution binary ICO.
c39abbc	style(footer): update footer logo badge container to white background	Replaced dark badge with clean white container and blue text.
219ce82	style(logo): update logo badge container and assets to clean white bg	Converted logo tagline to dark slate #0f172a for crisp contrast.
296f1f4	fix(player): convert video URLs to embed format and interactive locks	Resolved X-Frame-Options SAMEORIGIN YouTube embed errors.

5368e0f	fix(auth): persist session and view in localStorage across page refresh	Zero-logout browser refresh with instant state hydration.
5f106db	fix(admin): provide prominent pill toggles and inline restore buttons	Fixed category dropdown lock; added RotateCcw reset action.
46e0c4d	ci: replace fragile third-party FTP action with native Python FTP sync	Custom Python deploy engine with delta hash caching.
f37ded5	ci: use protocol ftp to prevent data socket reset on Hostinger	Eliminated passive data channel connection timeouts.
1b9b663	ci: auto-normalize FTP_SERVER_DIR to domains/aaroha-inc.com/...	Directory path auto-correction in deployment script.
ed0adce	ci: use npm install in workflow and sync package-lock.json	Clean CI dependency installation and cache normalization.
4d6bf9d	ci: trigger automated Hostinger deployment pipeline	First fully verified automatic CI/CD deployment execution.
a32e6de	ci: sanitize FTP server hostname and adjust default directory	Configured Hostinger production remote deployment target.
89c1781	ci: add GitHub Actions workflow for automated FTPS deployment	Initial continuous integration workflow pipeline creation.
a1ee2d6	ci: add production multi-stage Dockerfile and docker-compose	Coolify webhook and standalone container support.
942e43a	fix(ui): replace academic 'Curricula' terminology with 'Courses'	Coursera-calibrated navigation and smooth scroll behavior.
e1270f0	feat: complete Aaroha LMS enterprise platform baseline	Initial full-stack repository commit: React + PHP 8 + MySQL.

Production Verification Checklist	Verification Result
• Frontend Production Build: Clean Vite bundle generated in 513 ms without warnings	**PASS [513ms]**
• GitHub Actions CI/CD Pipeline: Workflow run 43222431909 completed successfully	**PASS [Success]**
• Live Hostinger Web Server: https://lms.aaroha-inc.com returning HTTP 200	**PASS [HTTP 200]**
• Favicon Distribution: /favicon.svg (23.1 KB) & /favicon.ico (36.8 KB) live	**PASS [200 OK]**
• Access Security Gate: Un-enrolled learners strictly restricted to Course Brochure	**ENFORCED [Gated]**